

Contactar

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Aptitudes principales

Artificial Intelligence (AI)
Generative AI
Retrieval-Augmented Generation
(RAG)

Languages

Español
Inglés

Certifications

Elastic Stack 7
Curso de Introducción al
Pensamiento Computacional con
Python
Curso de Docker
Curso de Organización y
Productividad con Notion
Curso de API REST

Honors-Awards

Evaluador BBVA Open Talent 2018

Oscar Giovanni Bocanegra Hurtado

Senior AI & Data Engineer | AI & Data Solutions Architect | GenAI,
RAG, AI Agents, Data Platforms, Microsoft Fabric

Área metropolitana de Bogotá D.C.

Extracto

I'm a Senior AI & Data Engineer and AI & Data Solutions Architect with 13+ years of experience designing and delivering enterprise software, Big Data, cloud data platforms, and AI solutions. I help companies, startups, scale-ups, technology consultancies, and enterprise teams design and build production-ready AI and Data solutions — from Generative AI, RAG and AI Agents to modern Data Platforms, Microsoft Fabric and intelligent automation. My work combines architecture-level thinking with hands-on engineering. I can define target architectures, integration patterns, data and AI platforms, governance requirements and technology stacks, and then work directly with engineering teams to implement and operationalize them. My experience includes:

- Generative AI, LLM applications, RAG and AI Agents
- Python, Spark and PySpark
- Data Engineering, ETL/ELT, Data Lakes and Lakehouse architectures
- Microsoft Fabric, OneLake and Microsoft Purview
- Data Hub and Data Mesh architectures
- AWS services including S3, Glue, Athena, Lambda, Step Functions, Secrets Manager, CloudWatch and API Gateway
- Data Governance, Solution Architecture and Cloud Architecture
- Self-hosted AI and Big Data platforms

I have worked on initiatives involving Microsoft Fabric-based Data Hub and Data Mesh architectures, cloud and data platform modernization, distributed data processing, AI/GenAI adoption, and enterprise-grade integration. Selected validated outcomes from my professional experience include:

- Reduced delivery times by up to 25%
- Automated up to 40% of previously manual tasks
- Improved KPI accuracy by up to 30%
- Reduced operational costs by up to 30%
- Improved operational efficiency by up to 20%

Core services:

- Enterprise RAG Solutions
- AI Agents & Automation
- Data Platform Engineering
- Microsoft Fabric Architecture
- Fractional AI/Data Architecture

I'm available for fixed-scope projects, contract engagements, and fractional AI/Data architecture work. If your

team needs senior AI/Data architecture combined with hands-on implementation, feel free to connect or message me.

Experiencia

SMX Services & Consulting, Inc.

Senior Data & AI Engineer

enero de 2024 - Present (2 años 9 meses)

Estados Unidos

Lead data-driven transformation initiatives by designing and implementing modern Data and AI architectures across Microsoft Fabric, OneLake, Lakehouse, Data Hub, and Data Mesh environments.

Key responsibilities and contributions:

- Design target architectures for enterprise Data Platforms using Microsoft Fabric, OneLake, Data Hub, Lakehouse, and Data Mesh patterns.
- Define standards and architectural guidelines for data ingestion, quality, governance, cataloging, lineage, traceability, and enterprise data consumption.
- Design Medallion Architecture patterns across Bronze, Silver, and Gold layers to improve data reliability, reuse, and availability.
- Develop and implement ETL/ELT pipelines and automation using Python, PySpark, APIs, and cloud services.
- Drive the adoption of Generative AI capabilities using Microsoft Foundry, LLMs, RAG, and enterprise AI Agents.
- Develop LLM-based prototypes for automated generation, validation, and documentation of data pipelines.
- Work with business and technology teams to identify, evaluate, and prioritize Data, Automation, and AI use cases.
- Lead migration initiatives from on-premise environments to AWS, focusing on scalability, maintainability, and operational efficiency.

Core technologies: Microsoft Fabric, OneLake, Microsoft Purview, Python, PySpark, Spark, ETL/ELT, APIs, Microsoft Foundry, LLMs, RAG, AI Agents, AWS.

StratioBD

Big Data Engineer

abril de 2022 - enero de 2024 (1 año 10 meses)

Колумбия

Worked as a Big Data Engineer within analytics initiatives for BanBajío in Mexico and Banco Internacional in Ecuador, delivering data-driven use cases and collaborating with technical teams and stakeholders.

- Led the design and delivery of data use cases and Proofs of Concept, reducing delivery times by 25% through process optimization.
- Designed and orchestrated data pipelines that automated 40% of previously manual tasks.
- Implemented AI models following CRISP-DM practices, improving KPI accuracy by 30%.
- Collaborated across engineering, analytics, and business stakeholders to translate requirements into scalable data solutions.

Consulting eSoluzion

Big Data Engineer

junio de 2021 - abril de 2022 (11 meses)

España

Worked as a Big Data Engineer supporting BBVA Spain, contributing to Data-Driven transformation initiatives focused on optimizing the data lifecycle across Cloud and Legacy environments.

- Led data transformation initiatives focused on improving the end-to-end data lifecycle.
- Integrated Cloud and Legacy technologies to support enterprise data modernization.
- Worked with distributed data processing and storage technologies including HDFS, Avro, Parquet, Spark, PySpark, and Python.

- Contributed to cloud-based data environments involving AWS and GCP.

Core technologies: HDFS, Avro, Parquet, Spark, PySpark, Python, AWS, GCP..

SETI S.A.S

Data Technical Team Leader

octubre de 2019 - mayo de 2021 (1 año 8 meses)

Bogotá

Led the implementation of a Data Lake for ATH Colombia, supporting the modernization of its data platform and improving data quality for strategic decision-making.

- Led Data Lake design and implementation activities aligned with the strategic objectives of the financial organization.
- Integrated internal and external data sources while establishing data quality and consistency practices.
- Contributed to a 30% reduction in operational costs through the implemented data platform and supporting processes.
- Led analytical initiatives including dashboards and predictive models to extract actionable insights.
- Coordinated technical teams and stakeholders while ensuring data quality, integrity, and regulatory compliance.

BBVA

Lead Software Development Engineer

agosto de 2017 - septiembre de 2019 (2 años 2 meses)

Colombia.

Led the implementation of BBVA Colombia's Document Management initiative, coordinating architecture, implementation, delivery, and multidisciplinary teams as part of the bank's digital transformation efforts.

- Led project planning, implementation, and technical delivery.

- Designed and implemented scalable solutions using MongoDB, Docker, Ansible, and Nuxeo.
- Applied Agile, traditional, and DevOps practices across the software delivery lifecycle.
- Coordinated multidisciplinary teams to align technical execution with strategic business objectives.
- Contributed to BBVA Colombia's broader digital transformation initiatives.

BBVA

Principal Software Development Engineer

agosto de 2014 - agosto de 2017 (3 años 1 mes)

Bogotá

Designed and optimized enterprise software solutions for BBVA Colombia, combining application engineering, databases, automation, and infrastructure technologies.

- Improved operational efficiency of key processes by 20% through software modernization and optimization.
- Worked across frontend, backend, relational and NoSQL technologies including Java, JavaScript, Oracle, SQL Server, MySQL, and MongoDB.
- Used Docker and Ansible to improve deployment automation and application delivery.
- Coordinated with multidisciplinary teams and implemented quality practices for enterprise-grade software delivery.

GLOKAL CONSULTING

Software Developer

abril de 2013 - agosto de 2014 (1 año 5 meses)

Bogotá

Developed and maintained business software applications, working across backend logic, databases, and application support. Collaborated with technical teams to implement functional requirements and resolve software issues.

ANTSS LTDA

Software Developer

abril de 2012 - diciembre de 2012 (9 meses)

Bogotá

Developed and supported software solutions for business operations, including application enhancements, database-related tasks, and technical troubleshooting. Contributed to the implementation and maintenance of enterprise applications.

WORLD OFFICE

IT Support Technician

marzo de 2009 - marzo de 2011 (2 años 1 mes)

Bogota

Provided technical support to end users, diagnosing and resolving hardware, software, connectivity, and application issues. Assisted with system configuration, user support, and day-to-day IT operations.

NOVAG INGENIERIA

IT Support Technician

junio de 2008 - febrero de 2009 (9 meses)

Bogota

Delivered end-user technical support, troubleshooting hardware, software, and network-related incidents. Supported installation, configuration, maintenance, and operational continuity of IT environments.

Bancolombia

Operations Assistant

febrero de 1999 - abril de 2007 (8 años 3 meses)

Colombia

Supported day-to-day operational and administrative processes within the banking environment, following established procedures and service standards. Assisted with transaction-related activities, operational controls, and internal process execution.

Educación

UNIR - La Universidad en Internet

Máster Universitario en Inteligencia Artificial, Artificial Intelligence · (febrero de 2023 - julio de 2025)

Instituto Europeo de Posgrado - IEP

Master Executive en Business Intelligence y Big Data, Tecnología en la gestión empresarial/Procesamiento de datos · (2019 - 2020)

Fundación Universitaria San Martín

Grado en Ingeniería, Gestión de proyectos de tecnología de la información · (2018 - 2018)

Fundación Universitaria San Martín

ingeniería de sistemas, Ingeniería informática · (2012 - 2018)

INSTITUTO SUPERIOR DE CARRERAS TECNICAS INSUTEC

Tecnico Profesional en Ingeniería de Sistemas, Facultad de Ingeniería de Sistemas · (1997 - 2000)